

PLA-1004 | Q BANK WITHOUT ANS

1. The tens digit of a two-digit number is two more than its unit digit. The two-digit number is 7 times the sum of the digits. Find the units digits?

- 3
- 1
- 4 ✓
- 2

2. Three partners shared the profit in a business in the ratio 5 : 7 : 8. They had partnered for 14 months, 8 months and 7 months respectively. What was the ratio of their investments?

- 38 : 28 : 21
- None of these
- 20 : 49 : 64 ✓
- 5 : 7 : 8

3. Find the remainder when 2321-2121 Divided by 2

- -1
- 2
- 1
- 0 ✓

4. The HCF of two numbers is 23 and the other two factors of their LCM are 13 and 14. The larger of the two numbers is:

- 299
- 276
- 322 ✓
- 345

$$23 \times 13$$
$$23 \times 14$$

5. The greatest number of five digits which is divisible by 32, 36, 40, 42 and 48 is

_____..

- 90725
- 90720 ✓
- 90715
- 90730

6. The difference between two numbers is 1146. When we divide the larger number by smaller we get 4 as quotient and 6 as remainder. Find the larger number

- 1485
- ☒ 1526
- 1432
- 1431

7. HCF of $\frac{3}{16}$, $\frac{5}{12}$, $\frac{7}{8}$ is:

- $\frac{5}{48}$
- $\frac{1}{48}$ ☒
- $\frac{3}{47}$
- $\frac{2}{47}$

8. The least number which when divided by 8, 12 and 16, leave in each remainder 3, is?

- 71
- 70
- 51 ☒
- 69

9. In ABC College, 63% of students are less than 20 years of age. The number of students more than 20 years of age is 4 more than $\frac{2}{3}$ of number of students of 20 years of age which is 42. What is the total number of students in the ABC College?

- ☒ 200
- 130
- 75
- 90

10. A man buys two cycles for a total cost of Rs. 900. By selling one for $\frac{4}{5}$ of its cost and other for $\frac{5}{4}$ of its cost, he makes a profit of Rs. 90 on the whole transaction. Find the cost price of lower priced cycle.

- Rs. 420
- Rs. 360
- ☒ Rs. 300
- Rs. 250

11. $25.25/2000$ is equal to _____.

- ☒ 0.12625
- 0.012625
- ☒ 1.012526
- 0.12526

12. A city has a population of 300000 out of which 180000 are males. 50% of the population are illiterate and if 70 % of the males are literate, then the number of literate females is:

- 16000
- 36000

- 2400
- 24000 ✓

13. Find the remainder when 421421 Divided by 7

- 6
- 1 ✓
- 2
- 4

14. The average score of a cricketer for ten matches is 38.9 runs. If the average for the first six matches is 42, then find the average for the last four matches.

- 33
- 37
- 31
- 34 ✓

15. Find the value in the place of question mark? $(26 \times 10 - 4 \times 17) / (14 \times 14 + 20)$

- 11|36
- 14|36
- 8|9
- 17|6

16. If the sales tax be reduced from $3\frac{1}{2}\%$ to $3\frac{1}{3}\%$ what difference does it make to a person who purchases article whose marked price is 4,200?

- 10
- 6
- 9
- 7

17. How many kg of rice, of cost 9 Rs/kg must be mixed with 27 kg of rice of cost 7 Rs/kg get a gain of 10 % by selling the mixture at 9.24 Rs/kg?

- 63
- 60
- 71
- 75

18. The average of three numbers x, y, and z is 30. If a fourth number w is added to the set, the average of the four numbers becomes 32. Find the value of w.

- 10
- 38
- 30
- 120

19. The average age of A and B is 24 years. The average age of B and C is 26 years, and the average age of A and C is 28 years. What is the age of B?

- 22 ✓
- 26
- 30
- 28

✗
↓

20. Find the value in the place of question mark? $(0.001 \times ?) / (200 \times 0.002) = 40$

- 8000000
- 16000
- 1600000 ✓
- None of these

21. Given : Three numbers 17, 42 and 93 Find the largest number to divide all the three numbers leaving the remainders 4, 3, and 15 respectively at the end?

- 10
- 12
- 13 ✓
- 14

22. There are two types of sugar. One is priced at Rs 62 per kg and the other is priced at Rs 72 per kg. If the two types are mixed together, the price of new mixture will be Rs 64.50 per kg. Find the ratio of the two types of sugar in this new mixture.

- 3:1 ✓
- 7:6
- 2:5
- 6:7

23. The average age of 40 students in a class is 15 years. The average age of the boys is 16 years, and the average age of the girls is 14 years. How many boys are there in the class?

- 20 ✓
- 25
- 15
- 30

24. In a group of 200 people, the average salary is \$50,000. The average salary of the men is \$55,000, while the average salary of the women is \$45,000. How many men are there in the group?

- 125
- 90
- 100 ✓
- 120

25. A and B invest in a business in the ratio 3 : 2. If 5% of the total profit goes to charity and A's share is Rs. 855, the total profit is:

- Rs. 1576

$$\frac{5x}{100} =$$

$$3x : 2x$$

$$2x = 855$$

- Rs. 1425
- Rs. 1537.50
- Rs. 1500 ✓

26. Which of the following fraction is smallest?

- $14/15$
- $21/24$
- $23/28$
- $15/19$ ✓

27. The product of two numbers is 4107. If the HCF of these numbers is 37, then the greater number is:

- 185
- 107
- 101
- 111 ✓

28. The least number which when 3 is added to it is completely divided by 8, 12 and 18 is?

- 69 ✓
- 71
- 70
- 58

29. P and Q entered into a partnership by investing Rs.1600 and Rs.1800 respectively and time period for P's investment is 3 months more than that for Q's investment. If profit share of Q is 78.75% of that of P, then find the time period for P's investment?

- 8 Months ✓
- 10 Months
- 9 Months
- 12 Months

30. Alpana started a business investing Rs. 45,000. After 3 months Priyanka joined her with a capital of Rs. 60,000. After another 6 months Kalpana joined them with a capital of Rs. 90,000. At the end of the year, they made a profit of Rs. 16,500. Find the share of Kalpana.

- Rs. 3800
- Rs. 2100
- Rs. 3300 ✓
- Rs. 3500

31. What percentage of numbers from 1 to 30 has 1 or 9 in the unit's digit?

- 15
- 121
- 22

- 20

32. A dishonest shopkeeper, at the time of selling and purchasing, weighs 10% less and 20% more per kilogram respectively. Find the percentage profit earned by treachery. (assuming he sells at Cost Price)

- 0.2
- 0.3
- 0.3333
- 0.25

33. The least square number which divides 8, 12 and 18 is?

- 144
- 64
- 100
- 121

34. A container contains 40 litres of milk. From this container 4 litres of milk was taken out and replaced by water. This process was repeated further two times. How much milk is now contained by the container?

- 28 litres
- 26.34 litres
- 27.36 litres
- 29.16 litres

35. A starts a business with Rs 'P' and 2 months later C joins him with Rs 'P/2', 1 month later B also joins him with Rs '3P'. The difference between the profit received by A and C at the end of one year is Rs. 9800. What is the difference between the profit received by B and C?

- Rs.30000
- Rs.32500
- Rs. 30800
- Rs. 28000

36. In a series of six consecutive even numbers, the sum of the second and sixth numbers is 24. What is the fourth number?

- 14
- 8
- 12
- 6

37. In $a : b = c : d$, b and c are called as _____.

- consequent
- mean
- extreme
- antecedent

38. If the number $517*324$ is completely divisible by 3, then the smallest whole number in the place of * will be:

- 2
- 1
- 0
- None of these

39. If $0.75 : x :: 5 : 8$, then x is equal to:

- 1.25
- 1.12
- 1.2
- 1.3

40. In ratio $a : b$, the first term a is called as _____.

- mean
- antecedent
- extreme
- consequent

41. If $M = y\%$ of z and $B = z\%$ of y , then which of the following must be true?

- M is more than N
- None of these
- Relation between M and N cannot be determined
- M is lesser than N

42. What is the unit digit of $973234! \times 234973!$?

- 7
- 6
- 2
- 9

43. Shankar sold a horse at a loss of 42.84%. How much percent should he gain in the next transaction so as to make an overall gain of 21.42%, if the two horses are bought at the same price?

- 0.6426
- 1
- 1.6426
- 0.8568

44. Two number are in the ratio 3 : 5. If 9 is subtracted from each, the new numbers are in the ratio 12 : 23. The smaller number is _____.

- 55
- 49
- 27

- 33

45. Find the value in the place of question mark? $\sqrt[3]{64} = 150$

- 1250
- 1200
- 2500
- 1500

46. A fruit seller always purchased mangoes in prime number. One particular day he purchased mangoes between 50-60 and gives 4 mangoes to his son. After selling the rest mangoes which was also above 50, he gained a profit which is equal to selling price of 4 mangoes. Find his profit percentage.

- 0.147
- 0.157
- 0.177
- 0.167

47. A pupil's marks were wrongly entered as 83 instead of 63. Due to that the average marks for the class got increased by half ($1/2$). The number of pupils in the class is:

- 40
- 70
- 20
- 10

48. Find the value of x: $\sqrt{x} = 0.02$

- 0.04
- 0.0004
- 0.004
- None of these

49. Six years ago, the ratio of the ages of Kunal and Sagar was 6 : 5. Four years hence, the ratio of their ages will be 11 : 10. What is Sagar's age at present?

- Cannot be determined
- 16 years
- 18 years
- 20 years

50. A car increases its speed by 25%. After that, it again increases its speed by 20%. By what percentage is the car's final speed greater than its original speed?

- 25
- 30
- 45
- 50

51. The price of sugar has increased by 20%. Anu has decided to only spend 8% more than she initially did on sugar. Find the percentage by which she needs to reduce the sugar consumption.

- 12
- 14
- 10
- 16

52. A tradesman gives a 4% discount on the marked price and gives an article free for buying every 15 articles and thus gains 35 %. find out by what percentage the marked price is above the cost price

- 0.406
- 0.5
- 0.396
- 0.22

53. Kavi and Bavi started a business with the investment of Rs.(x – 2000) and Rs.(x + 1000) respectively. After one year, Bavi withdrew 25% of her investment. At the end of two years, the total profit of the business is Rs.2400 and the profit share of A is Rs.1000. Find the value of x?

- 8000
- 7000
- 5000
- 6000

54. Ayesha's father was 38 years of age when she was born while her mother was 36 years old when her brother four years younger to her was born. What is the difference between the ages of her parents?

- 8 years
- 4 years
- 6 years
- 2 years

55. The ages of Ram and Anu are in the ratio 2 : 3. 15 years hence, the ratio of their ages will be 3 : 4. Find the present age of Anu.

- 45
- 15
- 30
- 60

56. If the number 46490# is exactly divisible by 72, then the minimum value of # is _____.

- 8
- 7

- 1
- 4

57. The greatest number of four digits which is divisible by 15, 25, 40 and 75 is:

- 9400
- 9600
- 9200
- 9800

58. The value of $(489.1375 \times 0.0483 \times 1.956)/(0.0873 \times 92.581 \times 99.749)$ is closest to

- 0.6
- 6
- 0.06
- 0.006

59. A man purchases 5 tables and 10 chairs for Rs.20,000. He sells the tables at 15% profit and the chairs at 10% loss. His overall profit is Rs.750. How much did he buy one table for

- Rs.2200
- Rs.900
- Rs.2000
- Rs.2600

60. A man sells an item for 700 and makes some profit. Then he sells the same item for 500 and his loss is $\frac{3}{5}$ th of the earlier profit. What is his approximate loss percentage in the second transaction?

- Data insufficient
- 11.5
- 15
- 13

61. Express 108.33...% as a fraction.

- 44990
- $\frac{13}{12}$
- $\frac{27}{25}$
- $\frac{108}{100}$

62. A man sells 5 articles for Rs.15 and makes a profit of 20%. Find his gain or loss percent if he sells 8 such articles for Rs.18.40.

- 8% loss
- 2.22% loss
- 2.22% profit
- 8% profit

63. Find the value in the place of question mark? $2002 + (200/200) = ?$

- 400001
- 4001
- 40001
- 20001

64. If $5.51 \times 10^k = 0.0551$, then the value of k is _____.

- -2
- -4
- -1
- -3

65. Find the remainder when 359421 Divided by 6

- 4
- 1
- 5
- 3

66. A man sells sugar to a tradesman at a profit of 20%, but tradesman, becoming bankrupt, pays only 72 paise per rupee. How much percent does the man gain or lose by his sale?

- 15% profit
- 12% profit
- 86.4% loss
- 13.6% loss

67. How much water is to be mixed in 36 litre of milk worth Rs. 5.00 per litre, so that value of mixture is Rs. 3.60 per litre?

- 12 litres
- 14 litres
- 10 litres
- 11 litres

68. The sum of three consecutive even numbers is 42. Find the middle number of the three?

- Rs. 14
- Rs. 24
- Rs. 16
- Rs. 18

69. 20% discount is offered on an item. By applying a promo code the customer wins 15% cash back. What is the effective discount?

- 12 percent
- 30 percent
- 32 percent...
- 30.8 percent...

70. Which one of the following numbers is exactly divisible by 11?

- 245642
- 235641
- 315624
- 415624

71. The average of 3, 8, 9 and x is 6 and the average of 17, 2, 5, x and y is 8. What is the value of y?

- 20
- 5
- 30
- 12

72. A dishonest milkman professes to sell his milk at cost price but he mixes it with water and thereby gains 25%. The percentage of water in the mixture is:

- 0.04
- 0.2
- 0.25
- 0.06

73. A man is 24 years older than his son. In two years, his age will be twice the age of his son. The present age of his son is _____.

- 18 years
- 22 years
- 20 years
- 14 years

74. When the price of an article was reduced by 20% its sale increased by 80%. What was the net effect on the revenue

- 44% Increase
- 66% Increase
- 66% decrease
- 44% decrease

75. Find the remainder when $2121+2321$ Divided by 11

- 2
- 1
- 3
- 0

76. If 40% of a number is equal to two-third of another number, what is the ratio of first number to the second number?

- 2:5
- 5:3

- 7:3
- 3:7

77. Two friends, Akash & Beenu had some candies each. One of them had 15 candies more than the other. The candies with Akash were 60% of the total candies with them. How many candies did each have?

- 40, 25
- 49, 34
- 47, 32
- 45, 30

78. The sum of three numbers is 98. If the ratio of the first to the second is 2 :3 and that of the second to the third is 5 : 8, then the second number is _____.

- 20
- 50
- 30
- 48

79. A company instead of raising the mark-up by 20% discounted the cost price by 20% while stitching the price tag on its product. Further the company offers a discount of 6.25% to its customer. In this process company incurs a loss of Rs. 37.5 on a single article. What is the selling price of that article?

- 365.5
- none of these
- 417.5
- 112.5

80. A merchant sells rice to a customer using false weights and gains 25% on selling at cost Price. What weight would a customer receive instead of a kilogram?

- 900 grams
- 850 grams
- 800 grams
- 750 grams

81. A shopkeeper gives a discount of 5%. If he gives a discount of 8%, then he earns Rs. 18 less. What is the marked price of the article?

- 550
- 700
- 500
- 600

82. Which of the following is largest fraction amongst $\frac{5}{8}$, $\frac{3}{7}$, $\frac{2}{9}$, $\frac{4}{5}$?

- $\frac{3}{7}$
- $\frac{5}{8}$
- $\frac{4}{5}$

- All are equal

83. A, B and C started the business with the investment in the ratio of 2:3:1. After 6 months, B left the business. At the end of the year, the total profit of the business is Rs.3600, then find the profit share of C?

- Rs.900
- Rs.1000
- Rs.800
- Rs.1200

84. The two given numbers A and B are in the ratio of 5:6 such that their LCM is 480. Find their HCF.

- 12
- 16
- 18
- 13

85. Karthi sells an article at a profit of 50%. If he had bought it at 40% less and sold it for Rs.21.50 less, He would have gained 60%. Find the cost price of the article.

- 56
- 52
- 48
- 40

86. P starts business with Rs.4500 and after 9 months, Q joins with P as his partner. After a year, the profit divided in the 3 : 5. What is Q's contribution in the capital?

- Rs.20,000
- Rs.10,000
- Rs.30,000
- Rs. 5,000

87. The HCF of two numbers is 11 and their LCM is 7700. If one of the number is 275, then the other number is:

- 283
- 318
- 308
- 279

88. A and B started a business with the investment of Rs.6000 and Rs.8000 respectively. After x months, A added Rs.2000 and B added Rs.1000 with their initial investment. At the end of one year, total profit is Rs.4800 and A's profit share is Rs.2200. Find the value of x.

- 3
- 8
- 6

- 4

89. Find the remainder when 363364 Divided by 7

- 4
- 1
- 2
- 3

90. A digital diary is sold for Rs.935 at a profit of 10%. What would have been the actual profit or loss on it, if it had been sold for Rs.810?

- Rs.50
- Rs.45
- Rs.40
- Rs.48

91. Find the average of the first 8 multiples of 7.

- 25
- 31.5
- 30
- 28

92. Four bells begin to toll together respectively at the intervals of 8, 10, 12 and 16 seconds. After how many seconds will they toll together again?

- 240 seconds
- 243 seconds
- 246 seconds
- 242 seconds

93. 16.67% of which amount is Rs 200?

- 1400 Rs
- 1200 Rs
- 1800 Rs
- 1500 Rs

94. The cost of 2 chairs and 3 tables is Rs.1300. The cost of 3 chairs and 2 tables is Rs.1200. The cost of each table is more than that of each chair by _____.

- Rs.100
- Rs.75
- Rs.70
- Rs.50

95. A student erroneously multiplied a number by $\frac{2}{5}$ instead of $\frac{5}{2}$. What is the percentage error in the calculation?

- 0.74

- 0.24
- 0.54
- 0.84

96. A jar full of whisky contains 40% alcohol. A part of this whisky is replaced by another containing 19% alcohol and now the percentage of alcohol was found to be 26%. The quantity of whisky replaced is:

- $\frac{1}{3}$
- $\frac{2}{3}$
- $\frac{2}{5}$
- $\frac{3}{5}$

97. The average of 9 numbers is 24. If the average of the first 4 numbers is 22 and the average of the last 4 numbers is 26, what is the 5th number?

- 32
- 36
- 24
- 20

98. Find the ratio in which rice at Rs. 7.20 a kg be mixed with rice at Rs. 5.70 a kg to produce a mixture worth Rs. 6.30 a kg

- 4:5
- 1:3
- 3:4
- 2:3

99. What is the remainder when $(1! + 2! + 3! + \dots + 100!)$ is divided by 5?

- 3
- 1
- 4
- 2

100. In ratio $a : b$, the second term b is called as _____.

- antecedent
- consequent
- extreme
- mean

101. A merchant marks his goods at Rs. 900 and allows a discount of 25%. If he still gain 12.5%, then the cost price of the article is:

- Rs. 720
- Rs. Cannot be determined
- Rs. 600
- Rs. 500

102. A bag contains a total of 93 coins in the form of one rupee and 50 paise coins. If the total value of coins in the bag is Rs.56, find the number of 50 paise coins in the bag?

- 74
- 70
- 64
- 60

103. The smallest 3 digit prime number is:

- 103
- 101
- 113
- 109

104. The fourth proportional to 5, 8, 15 is _____.

- 24
- 19
- 18
- 20

105. What is the unit digit of the sum of first 111 whole numbers?

- 6
- 4
- 5
- 0

106. Simplify: $13.05 \times 4.5 \div 0.5$

- 115.5
- 112.32
- 117.45
- 120.6

107. Find the remainder when 323232 Divided by 7

- 2
- 4 ✓
- 1
- 0

108. Three numbers are in the ratio of 3 : 4 : 5 and their L.C.M. is 2400. Their H.C.F. is:

- 50
- 40 ✓
- 70
- 30

109. Find the greatest number that will divide 43, 91 and 183 so as to leave the same remainder in each case.

- 3
- 7
- 5
- 4 ✓

110. Saransh is 50 years old and Nazma is 40 years old. How long ago was the ratio of their ages 3:2?

- 25 years
- ✓ 20 years
- 30 years
- 40 years

$$\frac{50-x}{40-x} = \frac{3}{2}$$

111. What is the fraction of decimal value 87.00013?(In Mixed fraction)

- 13/100087
- 87/100013
- 87 13/100000 ✓
- 13 87/100000

112. The price of an article is increased by 20% and then decreased by 20%. The final price as compared to the original price is:

- ✓ 4% less
- 4% more
- 20% less
- 20% more

LOSS ONLY

$$\therefore \left(\frac{x}{10}\right)^2 \Rightarrow \left(\frac{20}{100}\right)^2 = 4$$

113. The salaries A, B, C are in the ratio 2 : 3 : 5. If the increments of 15%, 10% and 20% are allowed respectively in their salaries, then what will be new ratio of their salaries?

- None of these
- Cannot be determined
- 3:3:10
- 10:11:20 ✓

114. FIND THE UNIT DIGIT OF 2284021

- 4
- 8
- 5
- 6 ✓

115. What is the unit digit in {(6374)1793×(625)317×(341491)}?

- 4
- 1

- 8
- 0

116. Sudhir and Tushar invested Rs. x and Rs. $3x$ respectively in a business. The time periods of Sudhir and Tushar is in the ratio 2 : 3. If the profit share of Tushar is Rs. 270000, then find the total profit if they continue the business for 3 more years?

- Rs. 330000
- Rs. 1320000
- Rs. 660000
- Rs. 1640000

117. A and B start a business. A invests Rs 'X' and B invests Rs 45000. 9 months later A withdraws half the amount. The profit received by A at the end of 1 year is 22.22% less than that of B. What is the value of 'X'?

- 43000
- 42000
- 40000
- 45000

118. A fruit seller had some oranges. He sells 30% oranges and still has 140 oranges. Originally, he had:

- 200 oranges
- 672 oranges
- 300 oranges
- 288 oranges

119. $A + B = 50$. Then A increases by 15 percent and B decreases by 20 percent. What is the value of B if $A - B = 38$ after the change?

- 5
- 10
- 8
- 40

120. The numerator of a certain fraction is 8 less than the denominator. If 3 is added to the numerator and 3 is subtracted from the denominator, the fraction becomes $\frac{3}{4}$. Find the original fraction?

- $\frac{2}{5}$
- $\frac{7}{9}$
- $\frac{3}{11}$
- $\frac{8}{5}$